

Who Bore Britain’s Trade Shocks?

A Household Anatomy of the Post-Covid Episodes*

Vahid Ahmadi[†]

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Abstract

Britain’s post-Covid decade delivered four major trade shocks and a placebo: the TCA border (2021), the sharpest terms-of-trade deterioration since the 1970s (2022), the US tariffs (2025), the India agreement (2026), and CPTPP accession. This paper studies them under one household lens—each episode a declared first stage from published estimates or government projections, mapped onto households through prices and earnings, compared in ratios because sizes span two orders of magnitude. First-pass results: the negative shocks were regressive (energy cost the bottom decile 12.39 per cent of spending versus 5.77 at the top; TCA food 1.19 versus 0.66), the projected positive gains are proportional and negligible, and the cushioning was margin-specific: automatic on earnings (36.6–43.9 per cent), discretionary on prices (the Energy Price Guarantee absorbed 46.2 per cent), with indexation a year late—£393 per Universal Credit household in the crisis year. A verified PolicyEngine microsimulation stage is specified.

Keywords: trade shocks, tariffs, Brexit, terms of trade, microsimulation, automatic stabilisers, public finances

JEL codes: F13, F14, F16, D31, H23, I38

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[†]Research Associate at PolicyEngine. Email: vahid@policyengine.org.

1 Introduction

Britain has just lived through the densest sequence of trade shocks in its post-war history: a new customs border with its largest trading partner (2021), the sharpest terms-of-trade deterioration since the 1970s (2022), a tariff war with its largest single-country export market (2025), and two new trade agreements (2024, 2026). Each episode has its own literature; none has been studied at the level where trade policy is actually felt—the household, through the tax–benefit system. This paper studies all of them there, together, under one lens.

The design holds the second stage fixed and varies the shock. Each episode enters as a *declared first stage*—published estimates for the negative shocks, the government’s own projections for the positive ones, never re-estimated here—and is mapped onto households through two channels: consumer prices against expenditure patterns, and labour income through the statutory tax–benefit system. Because gross sizes span two orders of magnitude, comparisons run in ratios: burden as a share of spending by income decile, cushioning shares, and per-pound-of-shock intensities. And because the rulebook itself moved across the window—the Covid uplift removed in 2021, upratings lagging a year behind prices, the Energy Price Guarantee legislated mid-shock—the design treats the tax–benefit vintage as an explicit dimension: the same shock is scored under the rules of its day, under the previous year’s rules, and with the discretionary layer switched off.

The first-pass results already discipline the received wisdom in three ways. First, the negative shocks were regressive and the positive ones were not progressive: the energy shock put 12.39 per cent of bottom-decile spending at stake against 5.77 per cent at the top, the TCA food-price effect 1.19 against 0.66 per cent, while the India agreement’s consumer gains are proportional and worth pounds-per-year, not per cent. Second, per pound of gross shock, the episodes differ in bite: the energy shock was the most bottom-decile-intensive (0.194 points of bottom-decile spending per £1 billion), the FTA gains the flattest (0.084). Third—the finding the rulebook axis exists for—the British state cushioned its trade shocks with different instruments on different margins: the earnings-led tariff shock was absorbed automatically (36.6 to 43.9 per cent, from the companion study), the price-led energy shock was absorbed by discretion (the Energy Price Guarantee took 46.2 per cent of the gross shock), and the automatic price-side mechanism—indexation—ran a year late, costing a Universal Credit household £393 (9.8 per cent of the allowance) in the crisis year, months after a £1,040 uplift had been withdrawn.

The contribution is threefold. First, comparability: no study, for any country, has run multiple trade shocks through a common statutory household lens; the closest traditions—the World Bank’s tariff incidence framework, the CGE-microsimulation linkages, the EUROMOD shock literature—each supply one ingredient (Section 2). Second, the rulebook axis: holding a shock fixed and varying the vintage of the welfare state that receives it has, to our knowledge, no precedent, and post-Covid Britain—four shocks landing on four rulebooks—is its natural laboratory. Third, honesty about epistemic status as a design principle: every first stage is labelled estimate or projection, every dial is declared, and one episode (CPTPP) is included purely as a placebo to demonstrate that the framework does not manufacture distributional findings from near-zero shocks.

The paper is a working draft in the same staged convention as its companion (Ahmadi, 2026): the first pass runs the declared first stages against published expenditure-by-decile tables and statutory arithmetic (Sections 5–6); the household-record stage—PolicyEngine UK on the enhanced Family Resources Survey, whose consumption imputation and parameter history have been verified for exactly this use—is specified in Section 7. Section 3 defines the estimand,

Section 4 the five episodes and their declared first stages, and Section 8 concludes.

2 Literature

Three literatures meet here without previously having met each other: the household incidence of trade policy, the microsimulation of shocks through statutory tax–benefit systems, and the UK’s episode-specific empirical record since 2020.

2.1 Trade to households

The framework for pushing trade shocks onto household microdata is best developed for developing countries: the World Bank’s Household Impacts of Tariffs programme simulates tariff changes through both expenditure and income portfolios for 54 countries (Artuç et al., 2021, 2019), and the OECD has published the concordance machinery for mapping CGE trade-model output onto household-budget-survey categories (Luu et al., 2020). The structural anchor for the consumption side is Fajgelbaum and Khandelwal (2016): trade’s gains flow disproportionately to poorer households through traded-goods budget shares. The 2025 tariff war produced a fast-moving US applied literature—the Budget Lab’s rolling distributional scoring of the enacted schedules and the Tax Policy Center’s implementation of tariffs inside a working tax microsimulation model (Budget Lab, 2025; TPC, 2025)—and a first academic treatment of two-sided incidence on household expenditure microdata (Fleck and Pradhan, 2026). The CGE–microsimulation linkage tradition supplies the architecture for putting a trade model upstairs and microdata downstairs (Robilliard and Robinson, 2005; Peichl, 2008; Colombo, 2010; Cali et al., 2019); of that tradition, only Peichl (2008) carries actual statutory tax rules in the micro layer.

What none of this work contains is a tax–benefit system responding to the shock. The World Bank and OECD frameworks stop at market incomes and expenditure; the US 2025 work is consumption-side; the CGE-linkage papers (Peichl’s flat-tax application aside) use reduced-form income equations. As far as we can establish, no trade shock has been run through a statutory tax–benefit calculator—EUROMOD, UKMOD, TAXBEN, PolicyEngine or any counterpart—in any country, for any episode.

2.2 Shocks through statutory rules

That machinery exists and has been applied to every kind of shock except trade. Dolls et al. (2012) measure the stabilisation coefficients of the US and European systems against stylised income and unemployment shocks; Brewer and Tasseva (2021) impose the observed Covid labour-market shock on FRS microdata through UKMOD and decompose the cushioning into automatic and discretionary parts; the EUROMOD nowcasting tradition supplies the labour-transition machinery (Navicke et al., 2014). The energy-price crisis produced the nearest price-side analogues: distributional analyses of domestic energy costs and the policy response by the IFS and Resolution Foundation, though none frames the imported-price component as a trade shock or reports stabilisation coefficients against it. The companion study (Ahmadi, 2026) is, to our knowledge, the first tax–benefit microsimulation of a trade shock anywhere—a single episode (the 2025 US tariffs), earnings channel only. This paper generalises it: four episodes, both channels, and the rulebook vintage as an explicit dimension.

2.3 The UK episode record

Each episode carries its own first-stage literature, reviewed with the imported numbers in Section 4: the Brexit/TCA price and trade effects (Bakker et al., 2023; Breinlich et al., 2022; Freeman et al., 2022), the 2022 terms-of-trade and energy-price shock and its policy response, the 2025 US tariff schedule (OBR, 2025), and the India agreement’s impact assessment (DBT, 2025). The distributional record of these episodes is thin everywhere and absent at the household-rulebook level: the Brexit price literature reports flat-to-regressive consumer-cost incidence but never asks what the benefit system gave back; the energy-crisis literature scores the discretionary response but not the trade shock as such; the FTA impact assessments contain, in DBT’s own regulator’s words, household analysis “developed further for future FTA IAs.” The gap this paper fills is the join: the same statutory second stage, run against all four first stages.

3 Framework: one estimand, four shocks, two channels, six rule-books

3.1 The estimand

For every episode the object is the same conditional quantity: the statutory household response to a declared trade-shock first stage. Let a first stage specify (i) a vector of consumer-price changes by consumption category, and/or (ii) a labour-income loss or gain with a sectoral location and an adjustment margin. The second stage maps that first stage onto households and computes the change in disposable income, its distribution, and the Exchequer’s share, under the tax–benefit rules in force at the episode date. The headline metric is the *cushioning rate*—one minus the household disposable-income change per pound of gross shock, the income-stabilisation coefficient of Dolls et al. (2012) generalised to price shocks by measuring the gross shock as the expenditure-weighted cost imposed at fixed consumption—because ratios, unlike levels, are comparable across episodes whose gross sizes span two orders of magnitude.

Nothing in the design estimates the first stages. Each is imported from published estimates and labelled with its source and its attribution caveats (Section 4); the paper’s contribution begins where those estimates stop, at the household and the Exchequer. This is the conditional-estimand convention of the companion tariff study (Ahmadi, 2026), applied retrospectively—with the advantage that retrospective first stages can be anchored to realised outturns rather than ex-ante bridges.

3.2 Two channels

Trade shocks reach households through prices and through earnings, and the post-Covid episodes split cleanly on that line. The consumption channel prices a category-level price vector against household expenditure shares: implemented at the household level on imputed consumption (the microsimulation stage) and, in this paper’s first-pass results, on published expenditure-by-decile tables. The earnings channel imposes sectoral labour-income changes on employees by industry and runs them through the statutory system, with the adjustment margin (wage cuts versus displacement) as a declared scenario dimension—the machinery, conventions and caveats of the companion study carry over unchanged. The channels are reported separately and never summed without an explicit interaction note: a price shock erodes real income that the benefit

system partially indexes, while an earnings shock triggers means-tested entitlement; the two responses run through different statutory mechanisms, and their comparison is the point.

3.3 The rulebook axis

The same shock cushions differently depending on when it lands, because the rulebook moves: benefit uprating follows lagged inflation, so a fast price shock outruns indexation for a year; thresholds frozen in cash terms tighten in real terms through an inflation episode; and discretionary interventions (most dramatically the Energy Price Guarantee) rewrite the effective schedule mid-shock. The design therefore separates three quantities for each episode: the cushioning delivered by the rules *as they stood*, the cushioning the same rules would have delivered under contemporaneous full indexation (the uprating-lag cost), and the discretionary layer on top. No study of trade shocks anywhere has, to our knowledge, held the shock fixed and varied the rulebook; the post-Covid sequence—four shocks landing on four different vintages of the same welfare state—is a natural laboratory for exactly that comparison.

3.4 What the design is not

The estimates are first-round statutory incidence, not general equilibrium: no behavioural response, no wage or employment feedback from the price shocks, no substitution in consumption (the price-channel numbers are first-order, envelope-theorem approximations to welfare changes and upper bounds on avoidable cost). Episode first stages carry their own literatures’ uncertainty, which the design brackets by importing low/central/high variants where the literature provides them and labelling single-point stages as such. The comparison across episodes is a comparison of conditional scenarios that share a second stage, not a causal decomposition of realised household outcomes over 2021–2026, which would require separating the trade shocks from everything else that happened—a task this design deliberately declines.

4 Five episodes and their declared first stages

Each episode enters the analysis as a declared first stage: published estimates, imported with citations and attribution caveats, never re-estimated. The five span the design’s axes—negative and positive, price-led and earnings-led, measured and projected—and one is included deliberately as a placebo.

4.1 Episode 1: the TCA border (2021–23), negative, price-led

The Trade and Cooperation Agreement took effect in January 2021, replacing single-market membership with a zero-tariff but non-tariff-barrier-laden border. The price evidence is the cleanest of any episode: exploiting cross-product variation in EU import intensity, [Bakker et al. \(2023\)](#) attribute roughly 8 percentage points of the 25 per cent food price rise between December 2019 and March 2023 to Brexit non-tariff barriers—a cumulative £250 per household on food alone, £6.95 billion in aggregate, with pass-through of NTB costs to consumers of 50–80 per cent and the largest effects in high-EU-import products such as meat and dairy (around 10 points above low-exposure comparators). The trade-volume record is equally stark: relative UK imports from the EU fell about 25 per cent on entry ([Freeman et al., 2022](#)), and by 2024 goods exports were an estimated £27 billion (6.4 per cent) below counterfactual, with some 16,400

small firms exiting EU exporting entirely (Freeman et al., 2024). The earnings side, by contrast, rests on assumptions rather than estimates—the OBR’s standing 15 per cent trade-intensity and 4 per cent productivity wedges (OBR, 2026)—so this episode’s earnings channel is declared as an assumption-based scenario and its price channel as an estimate-based one. Attribution caveat: the price identification is relative (cross-product), so economy-wide effects may be missed, and the counterfactual is contested by critics of the OBR’s trade assumptions.

4.2 Episode 2: the terms-of-trade shock (2022–23), negative, price-led, very large

The 2022 energy and import-price surge was, in the Bank of England’s description, a terms-of-trade shock of a size unseen since the mid-1970s: the terms of trade deteriorated 6.2 per cent in the year to 2022Q3 (ONS, 2023), import prices rose 20 per cent relative to output prices over two years (Broadbent, 2022), the energy import bill roughly doubled to £117 billion in 2022 (a transfer abroad of about 2.5 per cent of GDP—derived arithmetic, labelled as such), and of the roughly 17 per cent CPI rise to 2022Q4, more than 70 per cent (12.7 points) was imported (Dhingra and Page, 2023). The household-level price vector is policy-mediated and well documented: the typical annual energy bill’s cap rose from about £1,277 (winter 2021–22) to £3,549 (October 2022), truncated to £2,500 by the Energy Price Guarantee, at a discretionary cost the OBR put at £27 billion for the EPG within £47 billion of price subsidies and £78 billion of total energy support (OBR, 2022). Counterfactual no-policy welfare losses average 6 per cent of after-tax income, reaching 11 per cent at the 95th percentile of exposure (Levell et al., 2025), against energy budget shares of roughly 11 per cent in the second income decile versus 6 per cent in the ninth (Resolution Foundation, 2022). This is the strongest first stage of the five—multiple independent official estimates of both the aggregate shock and the retail price vector, plus a published imported/domestic decomposition—and the design decision it forces is constitutive for this paper: the shock is declared *gross*, and the Energy Price Guarantee is modelled as what it was, a discretionary rulebook intervention, so that automatic and discretionary cushioning can be separated.

4.3 Episode 3: the US tariffs (2025–26), negative, earnings-led

The April 2025 US schedule (10 per cent baseline, 25 per cent autos and steel) and its partial mitigation are the subject of the companion study (Ahmadi, 2026), which supplies this episode’s earnings first stage and its second-stage results: a tariff-calibrated earnings loss of roughly £886 million a year, cushioned at 43.9 per cent when delivered as broad wage cuts and 36.6 per cent as displacement. The external anchors are the OBR’s March 2025 scenarios (import-demand elasticity 0.4; peak GDP effects of -0.6 to -1.0 per cent across scenarios) (OBR, 2025) and the Bank of England’s effective-tariff estimates (11 per cent at April 2025, about 9 per cent after the Economic Prosperity Deal) (Bank of England, 2025). The UK consumer side of this episode is, to date, literally zero: the UK has not retaliated, so no tariff-induced consumer price vector exists—an empirical fact the comparative design records rather than an omission.

4.4 Episode 4: the India agreement (2026–), positive, both channels

The UK–India Comprehensive Economic and Trade Agreement entered into force on 15 July 2026. Its first stage is declared from the government’s own impact assessment (DBT, 2025): long-run GDP $+\text{£}4.8$ billion a year ($+0.13$ per cent by 2040), real wages $+0.19$ per cent ($\approx \text{£}2.2$

billion), and—on a different clock, at entry into force—consumer duty reductions of £220 million a year, £180 million of it on final goods, concentrated in clothing (£132 million), footwear (£13 million) and food and beverages (£12 million). Sectoral GVA effects give the earnings channel its geography: machinery and equipment +1.65 per cent, textiles and apparel −0.68 per cent. Two dials are declared rather than estimated, following the IA’s own caveats: preference utilisation (the duty figures assume 100 per cent on static 2022 trade) and consumer pass-through of the duty savings. The honest framing—and the reason a government projection can anchor a conditional design—is that the exercise computes what the government’s own central estimates imply for households, which its impact assessment does not report and its regulatory scrutineer asked for (RPC, 2026).

4.5 Episode 5: CPTPP accession (2024–), the placebo

The UK’s accession to the CPTPP (in force December 2024) carries a central government estimate of +£2.0 billion a year in the long run—0.08 per cent of GDP, about £70 per household per year before any incidence structure—because the UK already had agreements with nine of the eleven members (DBT, 2023). It is included precisely because it is nearly nothing: a comparative framework that can manufacture distributional findings from a near-zero shock is not to be trusted, and the CPTPP row is the demonstration that this one does not.

4.6 What the five episodes jointly reveal about the record

One asymmetry runs across the dossier and is itself a finding worth stating before any simulation: Britain’s *negative* trade shocks are measured—the TCA price effects and the terms-of-trade loss rest on published ex-post estimates—while its *positive* trade shocks are promised, resting on the government’s own ex-ante projections. The comparative design keeps that epistemic asymmetry explicit throughout: estimate-based and projection-based first stages are labelled as such in every table.

5 First-pass method

The results in this working draft run the declared first stages against public aggregates; the household-record stage is specified in Section 7. Everything is generated by one pipeline (`analysis_multishock/incidence.py`) that downloads and pins its raw inputs and emits every reported value as a machine-generated macro; no number is typed by hand except the declared first-stage constants of Section 4, which are typed *with their citations* and marked as imported.

Following the companion studies’ convention, the epistemic status of every input is stated once:

Object	Status
Episode first stages	Imported from cited published estimates (episodes 1–2) or government projections (episodes 4–5); episode 3 from the companion study. Never re-estimated here.
Expenditure by category $\times$ decile	Data (ONS Family Spending detailed workbooks).
Energy price-cap levels, statutory benefit rates, CPI outturns	Data (Ofgem/DESNZ, DWP, ONS).
Pass-through and utilisation dials	Assumed , bracketed, entering the pipeline once each.
Incidence arithmetic	Computed ; no estimation step anywhere in the pipeline.

Consumption incidence. Each episode’s price vector is applied to the relevant categories of household expenditure by equivalised-disposable-income decile, giving a per-decile burden or gain in pounds a year, as a share of total spending, and as a share of disposable income. These are first-order calculations: no substitution, no behavioural response, fixed baskets—upper bounds on avoidable welfare cost, in the envelope-theorem sense standard in this literature. Where a published aggregate exists (the £250 cumulative food cost of the TCA episode; the OBR’s Energy Price Guarantee cost), the pipeline reports its own aggregate beside it as a cross-check, not a calibration.

Rulebook arithmetic. The vintage axis is seeded with statutory arithmetic on public parameters: the real-terms shortfall of the Universal Credit standard allowance during the 2022–23 inflation episode relative to contemporaneous indexation (the uprating-lag cost, computed month by month against CPI outturns), and the discrete removal of the £20-a-week uplift in October 2021—the rulebook event that immediately preceded the terms-of-trade shock. The Energy Price Guarantee enters as the discretionary layer of episode 2: the gap between the gross price shock and the EPG-truncated shock is the discretionary cushioning share, computable per decile from cap levels alone.

Comparability. Episodes are compared in ratios, not levels: burden as a share of spending by decile, cushioning shares where computable, and a per-£-billion-of-shock normalisation, because gross sizes span more than two orders of magnitude and pound totals would reduce the comparison to a statement about the energy episode. Scenario dials (pass-through, utilisation) are reported as labelled variants, never blended into central cells; brackets are outer bounds, not confidence intervals; and estimate-based versus projection-based first stages are flagged in every table.

6 First-pass results

6.1 The comparative anatomy

Table 1 is the paper’s central object: the five episodes side by side under the common lens. Three regularities organise everything that follows. First, *Britain’s negative trade shocks were regressive and its positive ones were not progressive*: the energy shock cost the bottom decile 12.39 per cent of its spending against 5.77 per cent at the top, the TCA food shock 1.19 against 0.66 per cent, while the India agreement’s consumer gains are spread almost exactly proportionally

(0.015 versus 0.015 per cent of spending) and are two orders of magnitude smaller. Second, *intensity per pound of shock differs*: normalised per £1 billion of gross shock, the energy episode put 0.194 points of bottom-decile spending at stake, the TCA 0.162, and the CETA gains only 0.084—the composition of a trade shock, not just its size, determines who bears it. Third, *the cushioning came from different parts of the state*: the one earnings-led episode was cushioned automatically (36.6–43.9 per cent through the tax–benefit system, from the companion study), while the price-led energy episode was cushioned by discretion—the Energy Price Guarantee absorbed 46.2 per cent of the gross shock—and the automatic price-side mechanism, benefit indexation, arrived a year late (Section 6.3).

Table 1: Five trade shocks under one lens: first-pass comparability

Episode	Gross shock	Channel	% of spending		Per £bn (D1 pts/£bn)	Cushioning
			Decile 1	Decile 10		
TCA food NTBs (2021–23)	£7.37bn/yr	prices (food)	1.19	0.66	0.162	embedded in estimates
Energy terms-of-trade (2022–23)	£63.9bn/yr	prices (energy)	12.39	5.77	0.194	discretionary 46.2%
US tariffs (2025–26)	£886m/yr	earnings	consumer side zero		—	automatic 36.6–43.9%
India CETA (2026–)	£180.0m/yr (gain)	prices (clothing-led)	0.015	0.015	0.084	n/a
CPTPP (2024–), placebo	£2.0bn GDP	aggregate only	no structure claimed		—	n/a

Notes: first-order incidence on published expenditure-by-decile tables; fixed baskets, no substitution, no behavioural response. Episodes 1–2 rest on published estimates, episodes 4–5 on government projections, episode 3 on the companion study; the distinction is constitutive (Section 4). Deciles are of equivalised disposable income. The per-£bn column normalises bottom-decile spending exposure by gross shock size. Cushioning entries measure different mechanisms and are not summable.

6.2 Episode detail

TCA food prices. An 8 per cent NTB-attributed food price rise costs households £194 (bottom decile) to £315 (top decile) a year at end-state—regressive as a share of spending (1.19 versus 0.66 per cent) because food budget shares fall with income. The pipeline’s cumulative aggregate cross-check against the published £250 per household comes in at ratio 1.68 under a linear phase-in: the published estimate implies the price effect was back-loaded (an average effect about 0.30 of end-state over the window), which the first-pass linear ramp overstates. The diagnostic is reported, not calibrated away; the microsimulation stage will carry the published path.

The energy shock and the EPG. At cap levels, the gross price shock (£1,277 to £3,549 typical bill, +177.9 per cent) would have cost the bottom decile £2,017 a year—12.39 per cent of its total spending, more than double the top decile’s share. The Energy Price Guarantee truncated the rise to +95.8 per cent, absorbing £931 a year for a bottom-decile household and 46.2 per cent of the aggregate gross shock (£29.5 billion a year at the cap-level run-rate; £14.8 billion over the six-month EPG window, the same order as the OBR’s £27 billion costing). Because the EPG capped the *price*, its cushioning share is constant across deciles: a proportional subsidy on a regressive shock, so the post-EPG burden remained regressive—a design fact the household-record stage can contrast with targeted alternatives.

The US tariffs. The UK consumer side is zero—no retaliation occurred—so this episode’s incidence is entirely earnings-led: a gross loss of £886 million a year cushioned by the automatic

tax–benefit system at 36.6 per cent (displacement) to 43.9 per cent (wage cuts), the companion study’s headline carried over as this episode’s second stage.

The India gains, and the placebo. At full pass-through the CETA duty cuts are worth £2.96 a year to a bottom-decile household and £9.93 to a top-decile one—roughly proportional as a share of spending, and economically negligible at any pass-through dial (£1.48–4.97 at 50 per cent). The CPTPP placebo behaves as a placebo should: its £2.0 billion long-run GDP estimate divides to £70 per household per year with no defensible distributional structure, and the pipeline reports exactly that and nothing more.

6.3 The rulebook seed results

The vintage axis is seeded with two statutory facts. First, the uprating lag: Universal Credit’s standard allowance rose 3.1 per cent in April 2022 (indexed to the previous September’s inflation) while prices rose at double digits, so a single adult on UC fell £393 behind contemporaneous indexation over 2022–23—9.8 per cent of the annual allowance, lost to timing alone, at exactly the moment the terms-of-trade shock peaked. Second, sequencing: the £1,040-a-year Covid uplift was removed in October 2021, months before the energy shock landed, so the lowest-income households entered the largest price-led trade shock of the period from a rulebook trough. Together with the EPG’s 46-per-cent discretionary absorption, the seed picture of the British response to the 2020s trade shocks is: *automatic on the earnings margin, discretionary on the price margin, and lagged precisely where the price shocks bit hardest*. The microsimulation stage turns these representative-household facts into population-weighted decompositions.

7 The microsimulation stage

The first-pass results of Section 6 run the declared first stages against published expenditure-by-decile tables and representative-worker statutory arithmetic. The full stage runs them inside PolicyEngine UK on the enhanced Family Resources Survey, and four things change.

First, the consumption channel moves from decile tables to household records: the enhanced FRS carries imputed household consumption across the twelve COICOP divisions plus the sub-items that matter most for trade shocks (petrol and diesel separately, in pounds and litres; electricity and gas separately; bus fares), imputed by quantile regression forest from the LCFS 2023–24 and calibrated to administrative energy and fuel totals. Each episode’s price vector therefore prices each household’s own basket, jointly with that household’s benefit entitlements—the covariance between exposure and cushioning that decile tables cannot see. Tariff-line and category-level first stages must be pre-aggregated to those roughly sixteen categories, and the imputation freezes budget-share *patterns* at the 2023–24 donor vintage across simulated years, a limitation the stage will state. Second, the earnings channels run through the companion study’s displacement machinery on the same households, so the two channels of a given episode land on a common population and their incidence can be compared household-by-household rather than distribution-by-distribution. Third, the rulebook axis becomes exact: PolicyEngine’s parameter history carries dated, legislatively cited values through the whole window (the Covid-era Universal Credit uplift and its October 2021 removal, the March 2022 fuel-duty cut, the 2022 and 2023 upratings), its own test suite exercises the 2020–2023 rule years, and counterfactual-rulebook runs—an earlier year’s parameters applied to a later survey vintage—are native to the

architecture. Each episode is therefore scored under the rules in force at its date, under the preceding year’s rules (isolating uprating lag), and with the discretionary interventions switched off (isolating the automatic component)—the three-way decomposition of Section 3 computed rather than approximated. Two verified limits shape the earlier years: no 2021-vintage enhanced-FRS data year exists (the earliest survey base is 2022–23), so 2021 rulebooks are applied to later households with incomes re-deflated; and the survey’s Universal Credit/legacy-benefit split follows reported receipt rather than the historical rollout, so 2021–22 rule-year results understate legacy-benefit interactions and are restricted to the tax and UC-parameter channels. Fourth, the poverty and inequality outputs gain their correct household equivalisation and the Exchequer effects their full statutory coverage.

The build order follows the data dependencies. The consumption-channel implementation extends PolicyEngine’s existing consumption variables (used today for carbon and VAT analysis) with the episode price vectors—a parameter reform per episode, no new imputation. The earnings channels are re-runs of existing machinery with episode calibrations. The rulebook runs require only parameter-year switches plus explicit flags for the discretionary layers (the Energy Price Guarantee being the largest). The licensed inputs are the same FRS/EFRS files the companion study uses; nothing else is gated.

Two honest limits will survive the full stage. The consumption imputation carries its own error, inherited from the LCFS donor models, and imputation error correlated with benefit status would bias the exposure-cushioning covariance; the stage will report the imputation diagnostics alongside the results. And the rulebook decomposition attributes to “discretion” whatever the parameter history records as legislated change, which bundles genuine shock response with policy that would have happened anyway—a labelling choice, stated as such, not an identification claim.

8 Discussion

Three findings survive every caveat the first pass carries. The negative trade shocks of the 2020s were regressive through prices, and the positive ones offer no offsetting progressivity—partly because they are small, partly because their gains arrive in categories (clothing, footwear) whose budget shares are flat across the distribution. The state’s cushioning was margin-specific: automatic where the shock hit earnings, discretionary where it hit prices—and the discretionary instrument, capping a price, cushioned proportionally rather than progressively, so the post-intervention burden of the largest shock remained regressive. And the automatic price-side stabiliser the welfare state does possess—indexation—is calibrated to a world where prices move slowly: against a fast terms-of-trade shock it delivered its help a year late, which for the poorest households was the difference between the rulebook cushioning the shock and amplifying it.

One asymmetry deserves to be stated as a finding about the policy record rather than the method: Britain’s negative trade shocks are *measured*—their first stages rest on published ex-post estimates—while its positive trade shocks are *promised*, resting on the government’s own ex-ante projections. A comparative household lens makes the asymmetry consequential: the measured negative episodes are large and regressive; the projected positive episodes are small and proportional. On the government’s own numbers, the distributional ledger of the 2020s trade policy portfolio is in deficit, and nothing in the projected gains repays the measured losses’ incidence.

The limits are those declared throughout. First stages are imported, with their identification

and their controversies; the incidence is first-order and static; the channels are not summed; the rulebook decomposition labels legislated change as discretion without asking what would have been legislated anyway. The first pass runs on decile tables and representative households, not on household records—the covariance between exposure and entitlement, the poverty and inequality effects, and the population-weighted rulebook decompositions all await the microsimulation stage, which the verification work reported in Section 7 shows is buildable with existing tools on existing data.

What the framework offers, finally, is a standing capability rather than a one-off study: every future trade shock—a retaliation schedule, the next agreement, the next terms-of-trade event—arrives as one more declared first stage for the same second stage. A country that intends to conduct an active trade policy while running a means-tested welfare state needs exactly this join, and as of this writing no institution, in Britain or elsewhere, maintains it.

Data and code availability

The first-stage evidence dossier, the costing/incidence pipeline, pinned raw downloads and generated result macros are available in the accompanying GitHub repository (<https://github.com/PolicyEngine/uk-trade-shock-study>, directory `analysis_multishock/`); an archived release with a persistent identifier will accompany publication. All inputs to the first-pass results are public (ONS expenditure and price statistics, published first-stage estimates, statutory parameters). The microsimulation stage additionally requires the licensed Family Resources Survey via the UK Data Service, which is not redistributed. Episode first stages are imported from cited published estimates and are not re-estimated here.

Declarations

The author reports no conflicts of interest. No external funding was received for this study. The project uses secondary, aggregate public statistics and published estimates; no new human-participant data were collected. The author is responsible for the analysis, interpretation, and any remaining errors.

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